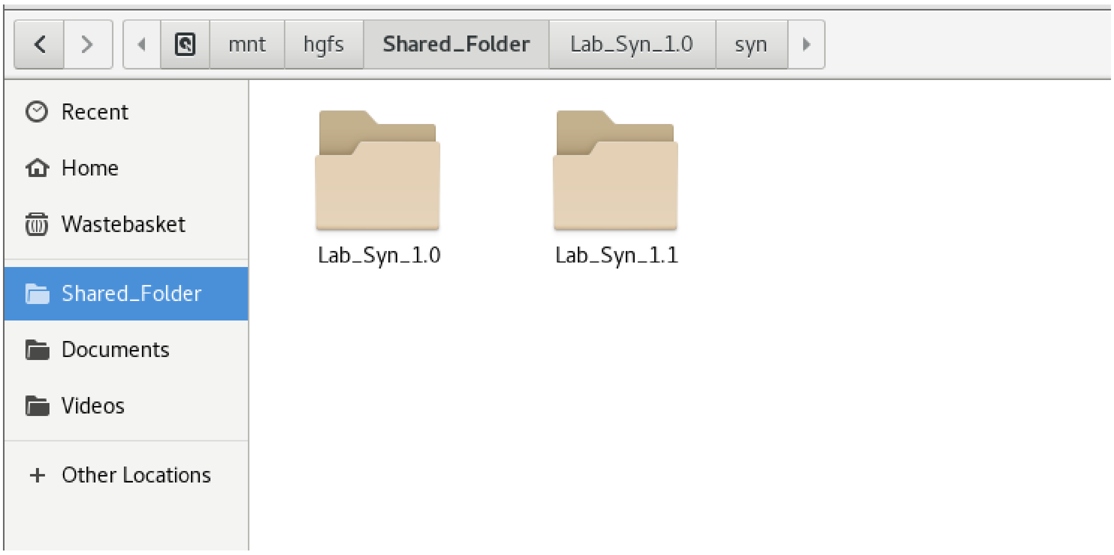
Lab_Syn_1.0

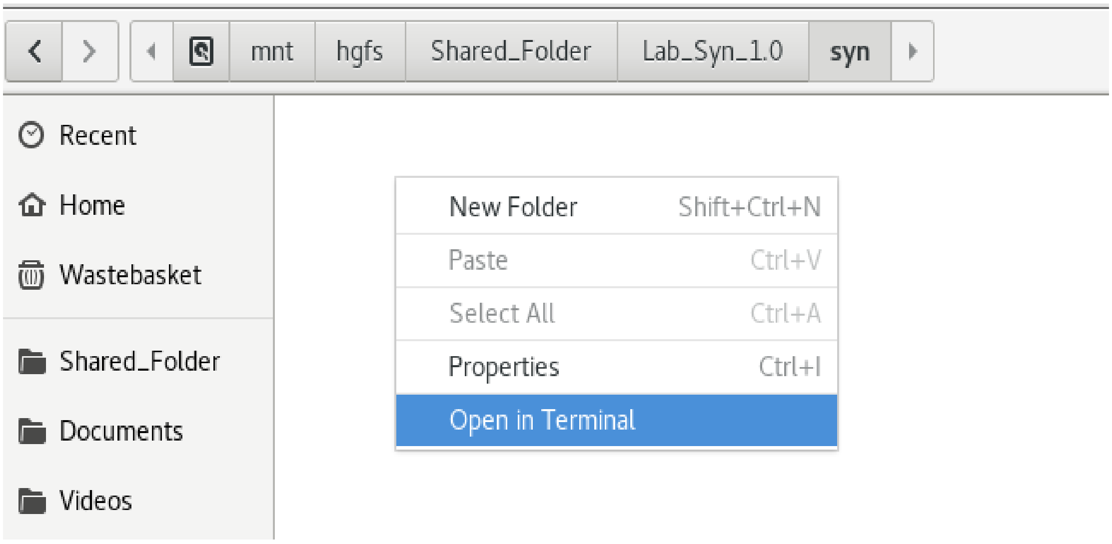
You have to synthesize Up_Dn_Counter.v file using Design Compiler GUI and generate Technology Dependent gate level netlists in Verilog format using standard cell library (scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db)

Steps:

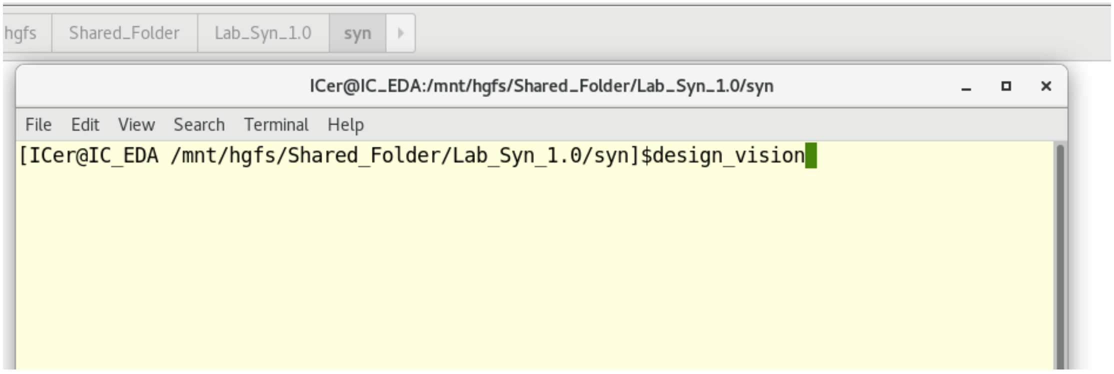
1. Place Lab_Syn_1.0 folder into your shared folder (Shared_Folder) in your computer. You will find Lab_Syn_1.0 appeared inside your virtual machine shared folder in this path (/mnt/hgfs/Shared_Folder).



2. Open the terminal inside path /mnt/hgfs/Shared_Folder/Lab_Syn_1.0/syn



3. Open design Compiler tool in GUI Mode by run design_vision command in the terminal



4. Load standard cell Library in db format
   1) Add standard cell libraries to target_library
   2) Add standard libraries to link_library
5. Reading Up_Dn_Counter.v rtl File Using Read Command

Once you are reading a RTL file, design compiler will do the followings: -

- Load GTECH library (To use it during translation of rtl to generic gates)
- Load the standard cell Libraries (To use in the mapping step)
- Checking HDL language syntax of the rtl files using PRESTO Compiler

```
design_vision> read_file -format verilog {/mnt/hgfs/Shared_Folder/Lab_Syn_1.0/rtl/Up_Dn_Counter.v}
Loading db file '/mnt/hgfs/Shared_Folder/Lab_Syn_1.1/std_cells/scmetro_tsmc_cl013g_rvt_tt_1p2v_25c.db'
Loading db file '/home/synopsys/syn/O-2018.06-SP1/libraries/syn/gtech.db'
Loading db file '/home/synopsys/syn/O-2018.06-SP1/libraries/syn/standard.sldb'
  Loading link library 'scmetro_tsmc_cl013g_rvt_tt_1p2v_25c'
  Loading link library 'gtech'
Loading verilog file '/mnt/hgfs/Shared_Folder/Lab_Syn_1.0/rtl/Up_Dn_Counter.v'
Detecting input file type automatically (-rtl or -netlist).
Reading with Presto HDL Compiler (equivalent to -rtl option).
Running PRESTO HDLC
Compiling source file /mnt/hgfs/Shared_Folder/Lab_Syn_1.0/rtl/Up_Dn_Counter.v

Inferred memory devices in process
        in routine Up_Dn_Counter line 14 in file
                '/mnt/hgfs/Shared_Folder/Lab_Syn_1.0/rtl/Up_Dn_Counter.v'.
===============================================================================
|    Register Name    |   Type    | Width | Bus | MB | AR | AS | SR | SS | ST |
===============================================================================
|     Counter_reg     | Flip-flop |   5   |  Y  | N  | N  | N  | N  | N  | N  |
===============================================================================
Presto compilation completed successfully.
Current design is now '/mnt/hgfs/Shared_Folder/Lab_Syn_1.0/rtl/Up_Dn_Counter.db:Up_Dn_Counter'
Loaded 1 design.
Current design is 'Up_Dn_Counter'.
design_vision>
Current design is 'Up_Dn_Counter'.
```

6. Link the design

7. **View design schematic in gtech cells**

8. Compile Up_Dn_Counter.v File Using Compile Command

9. View design schematic and find the inferred library cell name of Counter_reg[2] Flip Flop, its inputs and outputs

10. Generate the gate level netlist in Verilog format.

Repeat the Lab using Analyze/Elaborate instead of Read command